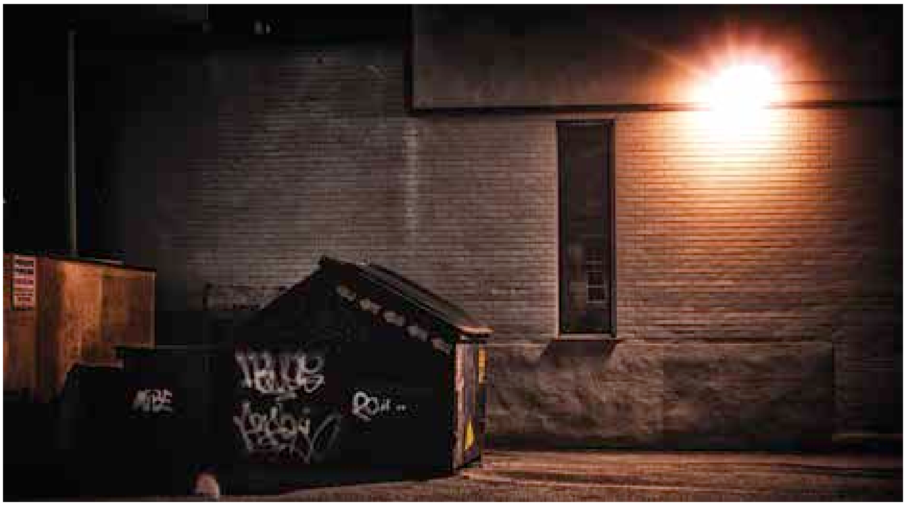

Where do we go now?

Back to the Present

Today's facts:

Here is where the storytelling ends and the science begins. The Hindu and Telegraph.co.uk reports that ethical permission for experimenting on cadavers has been granted to a collaboration between two research oriented biotechnology companies from different continents, who aim to eventually provide life-elongating services. While they certainly aren't the proverbial Umbrella Corp., this may just be the case someday. As it stands, the permission they needed was for performing experiments on the bodies of people who have recently been declared brain dead (but otherwise functionally half-alive). This trial, started in April 2016, will be the first milestone towards a potentially magical regenerative therapy that may one day help heal damaged brain and nervous tissue, not unlike a lizard that can grow back its tail. The companies mentioned in Telegraph's report were Revita Life Sciences in India, and Bioquark Inc. in the USA. Together they are collaborating on research towards the same end, in what is named the ReAnima project, in India. Yes, the zombie outbreak may actually start right here in India!! In the Rudrapur district of Uttarakhand, to be precise. A high population density certainly has its benefits!

While it's not a very large possibility that the ReAnima project will directly lead to a zombie apocalypse, it is definitely one of the first real possibilities, thanks to emerging biotechnologies. They describe their project as "A Second Chance at Life", and their mission statement is "Exploring the potential of cutting edge biomedical technology for human neuro-regeneration and neuro-reanimation". While the project is officially located in Philadelphia, in the USA, their first full scale trial is being conducted in the three-story Anupam Hospital in Rudrapur, India. This is most likely because technically, India has no laws for conducting clinical trials on the unfortunate 'living cadaver'. This is why they only required permission from an Institutional Review Board, regarding the ethical judgement of whether or not the study should be done. They will wait for patients from nearby hospitals who have been declared brain dead. The Principal Investigator of the study is Dr. Himanshu Bansal, an orthopedician with a passion for neuroscience, and the owner of Revita Life Sciences. The study is titled "Non-randomized, Open-labeled, Interventional, Single Group, Proof of Concept Study With Multi-modality Approach in Cases of Brain Death Due to Traumatic Brain Injury Having Diffuse Axonal Injury" and only recruits brain-dead humans. Before we look at what exactly the 'experiments' of this study are, and how zombies can come out of it, let's establish the characteristics of a zombie from the ground up.

The Science of Zombiefication

Thanks to media like books, and now movies, a lot of people have a general concept of what a zombie is. However, the zombies inhabiting different fictional worlds are often different in their natures, so here we are describing the possibilities of a zombie in our common real world. To be sure, there are many paths towards the possibility of a zombie outbreak, but the characteristics are mostly the same. The following information will help you analyze any zombie outbreak in the near future!

While the first human zombification may be the original practices that inspired the modern voodoo stories, it is not quite uncommon in nature as many insects are known to turn into zombies. Bees, crickets, and ants are infected by worms, fungi or other parasitic microorganisms that control their behaviour by hijacking their chemistry. The zombified hosts help propagate the controlling parasite by engaging in certain (often self-destructive) behaviour. In fact, a lot of us may already be zombies under the control of the protozoan Toxoplasma gondii, which is known to infect almost half the world's population.

Humans, however, have more complex systems in place than insects. Yet most of our functioning is controlled by the brain, both consciously and unconsciously. Therefore a zombie's brain is definitely different from a healthy human brain. But how? More often than not, zombies are brought back from the dead, and dead tissue atrophies. Thus, it would not be so far-fetched to explain a zombie's behaviour using a combination of the symptoms of various neurological problems and neurodegenerative diseases. Let us start with the most overt sign of a zombie, the movement. That stiff, uncertain, lumbering gait is a symptom of spinocerebellar ataxia, which is damage to the cerebellum, the smaller back of the brain that houses around half of all the neurons in the brain. This region functions like the firmware for our motor skills and coordination, so damage to the cerebellum is directly reflected in the retarded movements. Furthermore, their cognitive inability and impulsivity are most likely due to damage to the prefrontal cortex, the rational decision making part of